

Thermodynamic State Vector Inventory Discrepancy Evaluator: Formalizing Biogeochemical Mass Conservation in the Web of Life Architecture

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<https://github.com/pascalranoroarijaona/WebOfLife>

Sprint 61 Pre-Print

Abstract

Ecosystem simulations often suffer from numerical drift and mass-balance violations across complex biogeochemical cycles. Sprint 61 introduces the **Thermodynamic State Vector Inventory Discrepancy Evaluator** (`src/thermodynamics/state_validator.ts`), a robust verification module designed to enforce First and Second Law thermodynamic constraints within the Web of Life ecosystem architecture. By systematically comparing actual stock state deltas against integrated flux-derived expectations across Carbon, Nitrogen, Phosphorus, and Water pools, the `StateValidator` computes absolute discrepancies and evaluates global mass conservation under strict floating-point tolerances ($\tau = 10^{-6}$). This preprint outlines the mathematical foundations, architectural design, and verification suite for Sprint 61.

1 Introduction and Systems Ecology Context

In biophysical modeling, maintaining strict thermodynamic consistency is vital for long-term simulation stability. Ecosystems operate as open thermodynamic systems driven by external solar and radiative inputs, yet internal matter cycles (C, N, P, H_2O) must strictly obey mass conservation principles.

Sprint 61 addresses the challenge of inventory discrepancies—divergences between observed stock variations (ΔS_{actual}) and integrated boundary fluxes ($\Delta S_{\text{expected}}$)—by implementing the `StateValidator` component within the Web of Life framework.

2 Thermodynamic & Mass Balance Framework

2.1 First Law: Conservation of Mass

For any stock pool S_i over a discrete time step Δt , the actual stock delta is defined as:

$$\Delta S_{i,\text{actual}} = S_i(t + \Delta t) - S_i(t) \quad (1)$$

The expected flux-derived delta is computed via integrated boundary fluxes (inflows minus outflows):

$$\Delta S_{i,\text{expected}} = \Delta t \cdot \sum_j J_{ji} \quad (2)$$

where J_{ji} represents net flux rates entering or leaving stock i via pathway j .

2.2 Discrepancy Metrics and Tolerance

The absolute discrepancy ϵ_i for each stock key i is formalized as:

$$\epsilon_i = |\Delta S_{i,\text{actual}} - \Delta S_{i,\text{expected}}| \quad (3)$$

Total system absolute discrepancy E_{total} is aggregated across all elemental pools:

$$E_{\text{total}} = \sum_i \epsilon_i \quad (4)$$

Mass conservation is preserved if and only if:

$$E_{\text{total}} \leq \tau_{\text{tolerance}} \quad (5)$$

where default $\tau_{\text{tolerance}} = 10^{-6}$.

3 Architectural Design & Implementation

The `StateValidator` consumes `StateVector` instances (actual stock states) and `ThermodynamicStructure` models (expected flux transformations) to produce a structured `ValidationReport`.

```
export class StateValidator {
  private tolerance: number;

  constructor(tolerance: number = 1e-6) {
    this.tolerance = tolerance;
  }

  public evaluate(
    previousState: StateVector,
    currentState: StateVector,
    structure: ThermodynamicStructure,
    deltaTime: number
  ): ValidationReport {
    // Computes actual vs expected deltas and aggregates discrepancies
    ...
  }
}
```

4 Verification & Test Suite Specifications

The validation suite (`tests/sprint_061.test.ts`) tests three primary operational modes:

1. **Test 1 (Perfect Equilibrium / Conservation):** Verifies zero discrepancy when actual deltas perfectly match flux-derived expectations across C, N, P, and H_2O pools.
2. **Test 2 (Unmodeled Perturbation Detection):** Detects non-zero absolute discrepancies when unmodeled external forces perturb specific stock vectors.
3. **Test 3 (Tolerance Boundary Thresholding):** Confirms that the `isMassConserved` boolean correctly toggles based on whether total discrepancies breach $\tau = 10^{-6}$.

5 Conclusion

Sprint 61 establishes rigorous auditing mechanisms within the Web of Life framework, ensuring that all simulated biogeochemical cycles remain strictly bound by thermodynamic laws. Future sprints will leverage these validation reports to drive automated feedback-loop corrections in exergy dissipation pathways.

For complete implementation details and source code, visit the official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.